RESEARCH PROPOSAL PRESENTATION

Converting a Traditional Farm into an AI/ML-Powered Automated Smart Farm

Research Group

R26-IT-093

Presented by

Research Group Members

BACKGROUND

Introduction

Traditional livestock farming relies heavily on manual monitoring and decision-making.

Farmers face challenges in animal feeding, disease detection, detection, reproduction management, and market price fluctuations.

Artificial Intelligence (AI), Machine Learning (ML), and Internet of Things (IoT) technologies can improve farm management.

This research proposes an integrated smart farm system to improve productivity, efficiency, and animal welfare.

TRADITIONAL LIVESTOCK FARMING FACES SEVERAL CHALLENGES

Problem Statement

- Inconsistent feeding practices can affect animal nutrition and productivity.
- Delayed disease detection can increase treatment delays and health risks.
- Unpredictable animal reproduction cycles can lead to financial losses.
- Fluctuating market prices make profitable business decisions difficult.
- Limited real-time monitoring reduces the efficiency of farm management.

MAIN PROBLEM

The lack of an integrated, intelligent system for efficient livestock monitoring and data-driven farm management.

WHAT THE RESEARCH SETS OUT TO DO

Research Aim and Objectives

RESEARCH AIM

To develop an integrated AI/ML-powered smart farm system that improves livestock management through intelligent monitoring, prediction, and automation.

Research Objectives

1. Develop an automated feeding and personalized nutrition system.
2. Detect abnormal animal behavior and generate emergency alerts.
3. Predict healthy animal births and successful egg hatching.
4. Forecast livestock-related market prices using AI/ML.
5. Support farmers in making informed decisions using farm data.

FOUR MAIN COMPONENTS

Proposed Smart Farm Solution

Automated Feeding & Nutrition System

Recommends suitable feed and automates feeding based on animal requirements.

Disease Detection & Emergency Alerts

Identifies abnormal behavior and notifies farmers or veterinarians.

Animal Birth & Egg Hatch Prediction

Predicts birth or hatching outcomes using historical, image, and sensor data.

Customized Live Market Prediction

Forecasts market prices and and provides decision-support insights.

COMPONENT 1 OF 4

Automated Feeding & Nutrition

Key Functions

- Monitor animal behavior and feeding patterns.
- Identify animals and assess their physical condition.
- Analyze species, breed, weight, size, and health condition.
- Recommend suitable feed type, quantity, and feeding schedule.
- Automatically dispense the recommended amount of feed.

Expected Benefits

- Personalized nutrition.
- Reduced food wastage.
- Improved animal growth and productivity.
- More efficient farm management.

COMPONENT 2 OF 4

Disease Detection & Emergency Alerts

Key Functions

- Monitor animal movement, posture, and activity using cameras and sensors.
- Analyze behavioral changes using AI/ML.
- Identify possible signs of illness, such as limping and reduced activity.
- Generate alerts when abnormal conditions are detected.
- Recommend initial actions and notify veterinarians when necessary.

Expected Benefits

- Earlier identification of possible health problems. problems.
- Faster response to animal health issues.
- Reduced risk of disease spreading.
- Improved animal welfare.

COMPONENT 3 OF 4

Animal Birth & Egg Hatch Prediction

Key Functions

- Collect historical breeding records, sensor data, images, and environmental information.
- Analyze relevant patterns using AI/ML models.
- Predict the likelihood of a healthy calf birth or successful egg hatching.
- Generate early warnings when abnormal conditions are identified.
- Suggest preventive actions to improve outcomes.

Expected Benefits

- Early identification of potential risks.
- Improved preparation for births and hatching. hatching.
- Reduced livestock and poultry losses.
- Better farm productivity.

COMPONENT 4 OF 4

Customized Live Market Prediction

Key Functions

- Collect and update historical and available current market prices for milk, livestock, eggs, and meat.
- Analyze price patterns, seasonal changes, and demand-supply trends.
- Predict future market price movements using AI/ML.
- Provide market insights to support buying and selling decisions.
- Help farmers understand potential income, price changes, and financial risks.

Expected Benefits

- Better market awareness.
- Improved decision-making.
- Support for income planning.
- Reduced financial uncertainty.

COMPONENT 4 · INNOVATION

A customized livestock-market forecasting approach that combines market information with relevant farm production data and seasonal patterns.

HOW THE RESEARCH WILL BE CARRIED OUT

Data Collection & Methodology

Data Collection

- Collect relevant farm data from National Livestock Development Board (NLDB) farms.
- Use historical records related to animal health, breeding, production, and market prices.
- Gather available image and sensor data where applicable.
- Organize and validate the collected data.

Research Methodology

1. Data collection
2. Data cleaning and preprocessing
3. Feature selection and preparation
4. AI/ML model development
5. Model testing and evaluation
6. System integration and validation

WHAT THE SYSTEM SHOULD DELIVER

Expected Outcomes

- ✓ Improved livestock monitoring and management.
- ✓ More personalized and efficient animal feeding.
- ✓ Earlier detection of potential animal health problems.
- ✓ Better birth and egg-hatching risk prediction.
- ✓ Improved market price forecasting and decision support.
- ✓ Reduced avoidable losses and operational costs.
- ✓ More efficient and data-driven farm management.

RESEARCH CONTRIBUTION

An integrated smart farm approach combining four AI/ML components in one livestock management system.

SUMMARY

Conclusion

01

The proposed smart farm system addresses key challenges in traditional livestock farming.

02

AI/ML technologies support intelligent feeding, health monitoring, reproduction prediction, and market forecasting. forecasting.

03

The integration of these components aims to improve farm efficiency, animal welfare, and decision-making.

04

The system provides a foundation for more intelligent and and sustainable livestock farming.

Thank You

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